Contraction mappings

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Let $T: S \to S$ be a contraction mapping with a contraction parameter α in some norm:

$$\| T(x) - T(y) \| \leq \alpha \|x - y\| ,$$

where $S \subseteq \mathbb{R}^n$ is closed. A norm $\|\cdot\|$ is any function that satisfies

(a) $\|x+y\| \leq \|x\| + \|y\|$ (triangle inequality)

(b) $\|x\| = 0$ iff $x = 0$

(c) $\|ax\| = |a| \|x\|$, for any $a \in \mathbb{R}$.

□

Examples of norms:

(i) $\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$

(ii) $\|x\|_p = \left(|x_1|^p + \cdots + |x_n|^p\right)^{1/n}$, $1 \leq p < \infty$

(iii) $\|x\|_\infty = \max_i |x_i|$

(iv) $\|x\|_{p,w} = \left(\sum_i w_i |x_i|^p\right)^{1/p}$ for a vector w with all >0 entries

(v) $\|x\|_{\infty, w} = \max_i w_i |x_i|$; $w_i > 0 \ \forall i$

(vi) $\|x\|_{2,M} = (x^T M x)^{1/2}$ where M is a positive definite matrix

□

Theorem: $\exists$ a unique x^* s.t. $T(x^*) = x^*$. Such an x^* is called a fixed point. Further, x^* can be obtained as the limit of the iteration $x_{k+1} = T(x_k) \quad \forall x_0 \in S$

Proof: We will show that $\{x_k\}$ form a Cauchy sequence:

$$\|x_{k+1} - x_k\| \leq \|T(x_k) - T(x_{k-1})\|$$

$$\leq \alpha \|x_k - x_{k-1}\|$$

$$\Rightarrow \|x_{n+1} - x_n\| \leq \alpha^n \|x_1 - x_0\|$$

$\forall \ell \geq 1$

$$\Rightarrow \|x_{n+\ell} - x_n\| \leq \sum_{j=1}^{\ell} \|x_{n+j} - x_{n-j+1}\|$$

$$\leq \sum_{j=1}^{\ell} \alpha^{n+j} \|x_1 - x_0\|$$

$$= \alpha^n \left(\sum_{j=1}^{\ell} \alpha^j\right) \|x_1 - x_0\|$$

$$\leq \frac{\alpha^n}{1-\alpha} \|x_1 - x_0\| \quad — ①$$

$\Rightarrow \{x_n\}$ is a Cauchy sequence and since S is closed, $x_n \to$ some $x^* \in S$.

Since a contraction mapping is continuous and

$$x_{k+1} = T(x_k)$$

$$\Rightarrow \lim_{k\to\infty} x_{k+1} = \lim_{k\to\infty} T(x_k) = T\left(\lim_{k\to\infty} x_k\right)$$

↑ continuity of T

$$\Rightarrow \boxed{x^* = T(x^*)}$$

$\Rightarrow x^*$ is a fixed point.

Finally, we want to prove the uniqueness of the fixed point. Suppose x^* and y^* are two fixed points, then

$$\|x^* - y^*\| = \|T(x^*) - T(y^*)\|$$
$$\leq \alpha \|x^* - y^*\|$$

$$\Rightarrow \|x^* - y^*\| = 0 \quad \Rightarrow \quad x^* = y^*$$

□

Note: from ①, by letting $\ell \to \infty$, we have

$$\boxed{\|x_n - x^*\| \leq \frac{\alpha^n}{1-\alpha} \|x_1 - x_0\|}$$

($\|\cdot\|$ is a continuous function)

Note: While we will not need it for this course, the contraction mapping theorem holds much more generally, in non-empty complete metric spaces.

Cauchy sequences: A sequence $\{x_n\}$ is Cauchy if

$$\lim_{n\to\infty} \sup_{m \geq 0} \|x_{m+n} - x_n\| = 0$$

□

Fact: If a sequence is Cauchy in $\mathbb{R}^n$, then it has a limit x^*, i.e.,

$$\lim_{n\to\infty} x_n = x^*$$

Further if $\{x_n\} \in$ closed set, then $x^* \in$ closed set.

The usefulness of Cauchy sequences is due to the fact that you can determine convergence without knowing the limit x^*.